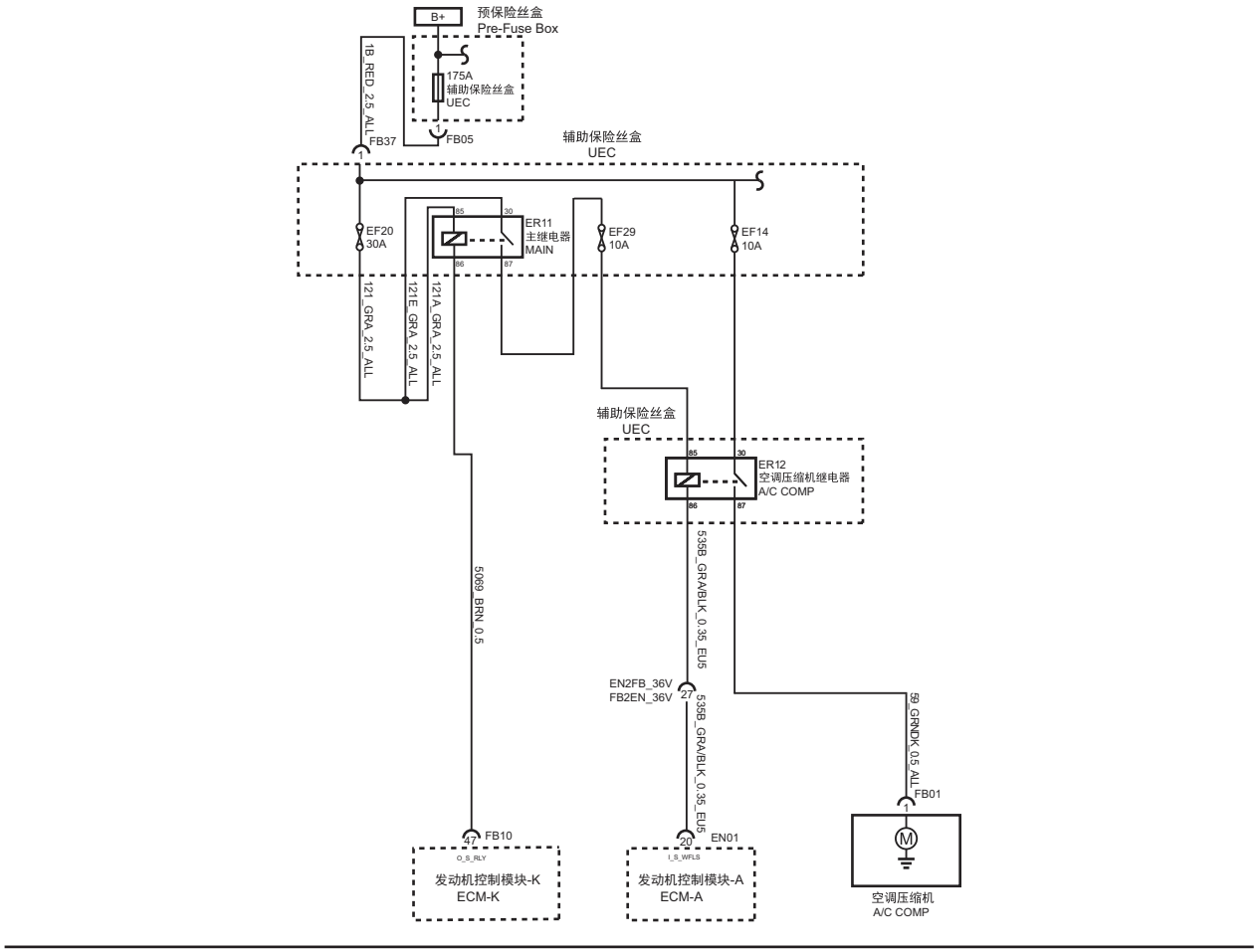
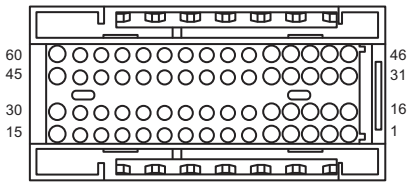
P0645、 P0646、P0647

Description of fault diagnosis codes
P0645-13：No load error on power stage for the compressor
P0645-4B：Over temperature error on powerstage for the compressor
P0646-11：Short circuit to ground error on power stage for the compressor
P0647-F0：Short circuit to battery error on power stage for the compressor
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Line or harness plug fault • Compressor relay fault
Description of circuit diagram
The engine control mode will drive the A/C compressor through controlling the A/C compressor relay. ECM detects drive circuit fault of A/C compressor.

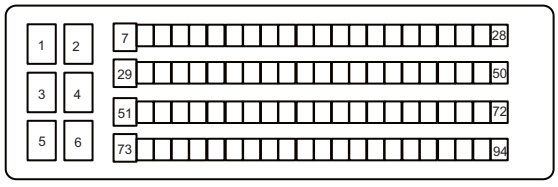
Circuit diagram



EN01



FB10



SV610856

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse EF29 for damage. 2 Check battery capacity. 3 Check the harness plugs of engine control module and air condition compressor relay for damage, poor contact, aging and looseness. 4 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON. 3 Diagnose the engine control system with diagnostic apparatus. 4 Are there any other fault codes except for P0645, P0646 and P0647?
		Yes → Refer to:DTC Summary List (EU5).
No ↓		
Step 3	Check the air conditioning compressor relay.	
		1 Turn the ignition switch to OFF. 2 Unplug the compressor relay. 3 Measure the resistance between 85 and 30,85 and 87,86 and 30,86 and 87 of compressor relay. **Standard value: ∞** 4 Apply battery voltage on the clearance between terminal 85 and terminal 86 of air condition compressor relay. 5 Measure the resistance between No.30 terminal and No.87 terminal of compressor relay. **Standard value: ≈ 0Ω** 6 Check whether the result is normal?
		No → Replace the air condition compressor relay.
Yes ↓		

Step 4	Check the power supply circuit of air condition compressor relay.
	1 Turn the ignition switch to OFF. 2 Unplug the compressor relay. 3 Measure the voltage between terminal of compressor relay and ground. Measuring circuit: the voltage between terminal 85 of compressor relay and ground. **Standard value: 10~14V** 4 Check whether the result is normal?
	No → Go to Step 6.
Yes ↓	
Step 5	Check the continuity of harness of air condition compressor relay.
	1 Turn the ignition switch to OFF. 2 Unplug the compressor relay. 3 Disconnect the ECM harness plug. 4 Measure the resistance between terminal of compressor relay and ECM harness terminal. Measuring circuit: the resistance between terminal 86 of compressor relay and No.20 terminal of ECM harness plug. **Standard value: ≈ 0Ω** 5 Check whether the result is normal?
	No → Repair or replace the faulted parts
Yes ↓	
Step 6	Check whether the harness of air condition compressor relay has grounding short circuit.
	1 Turn the ignition switch to OFF. 2 Unplug the main relay. 3 Unplug the compressor relay. 4 Disconnect the ECM harness plug. 5 Measure the resistance between terminal of air compressor relay and ground. Measuring circuit: the resistance between terminal 85 of air condition compressor relay and ground. Measuring circuit: the resistance between terminal 86 of air condition compressor relay and ground. **Standard value: ∞** 6 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 7	Check the engine control module.	
		1 Replace engine control module.
Next ↓		
Step 8	Complete the check and confirm whether the system works normally nor not.	